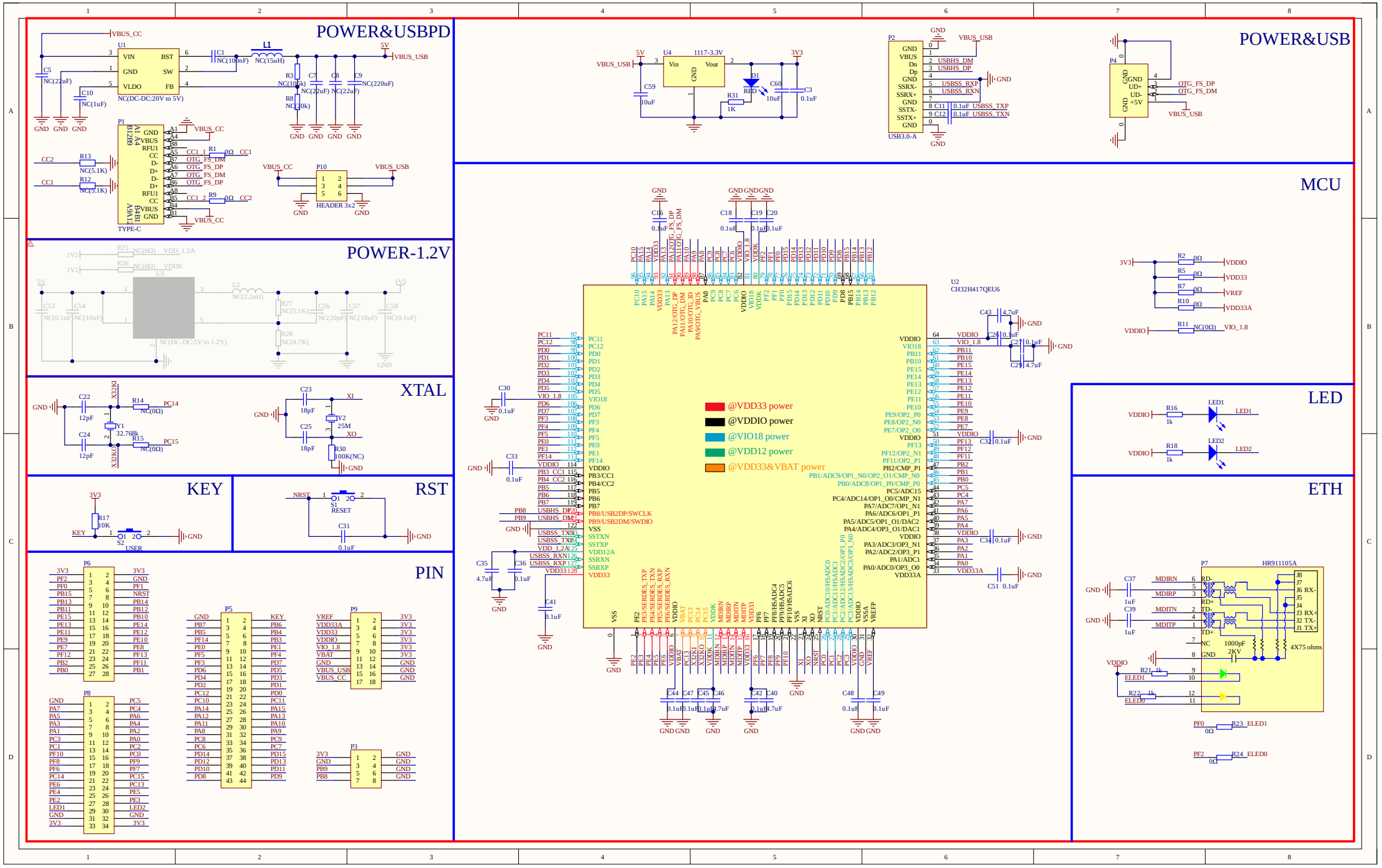
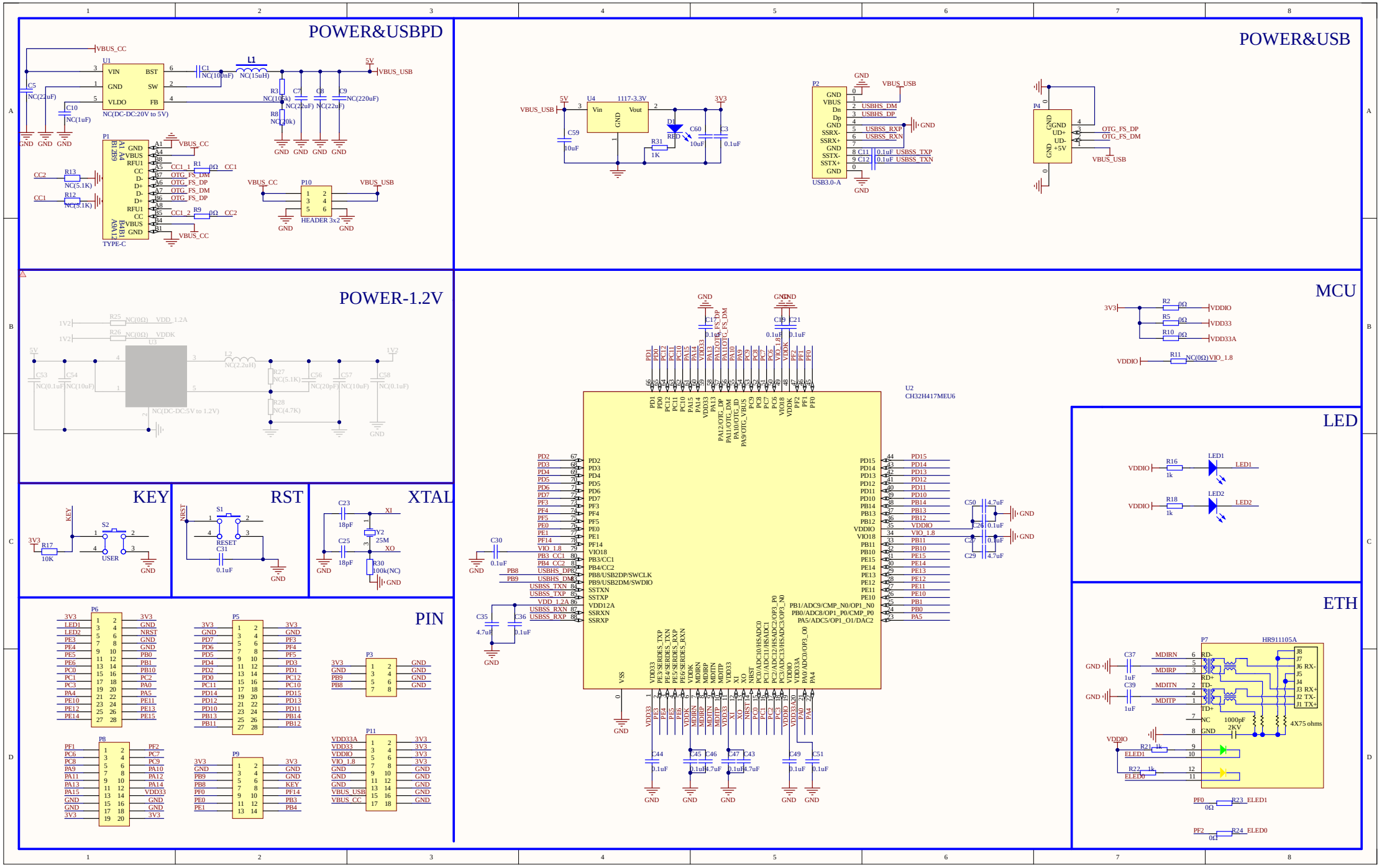
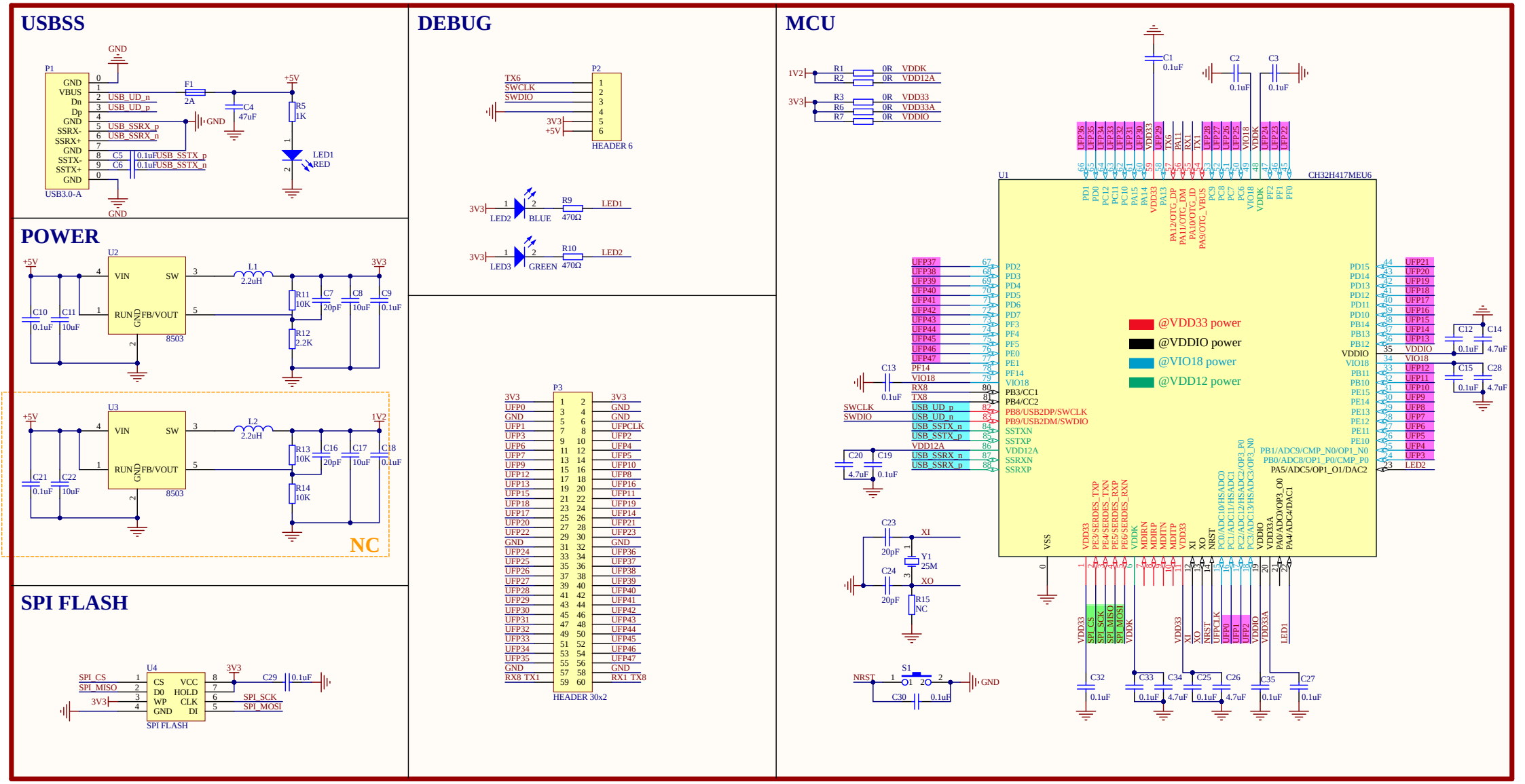
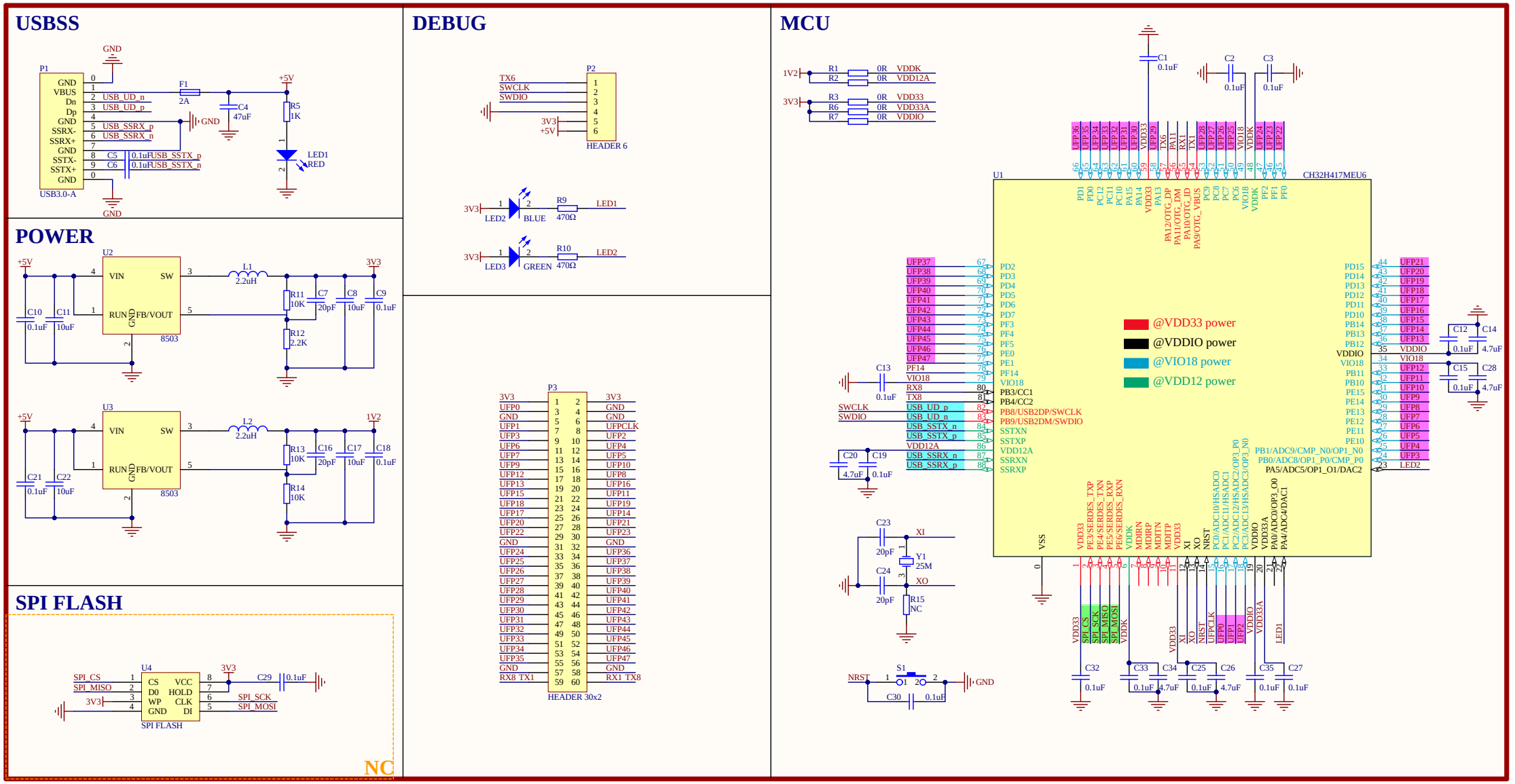
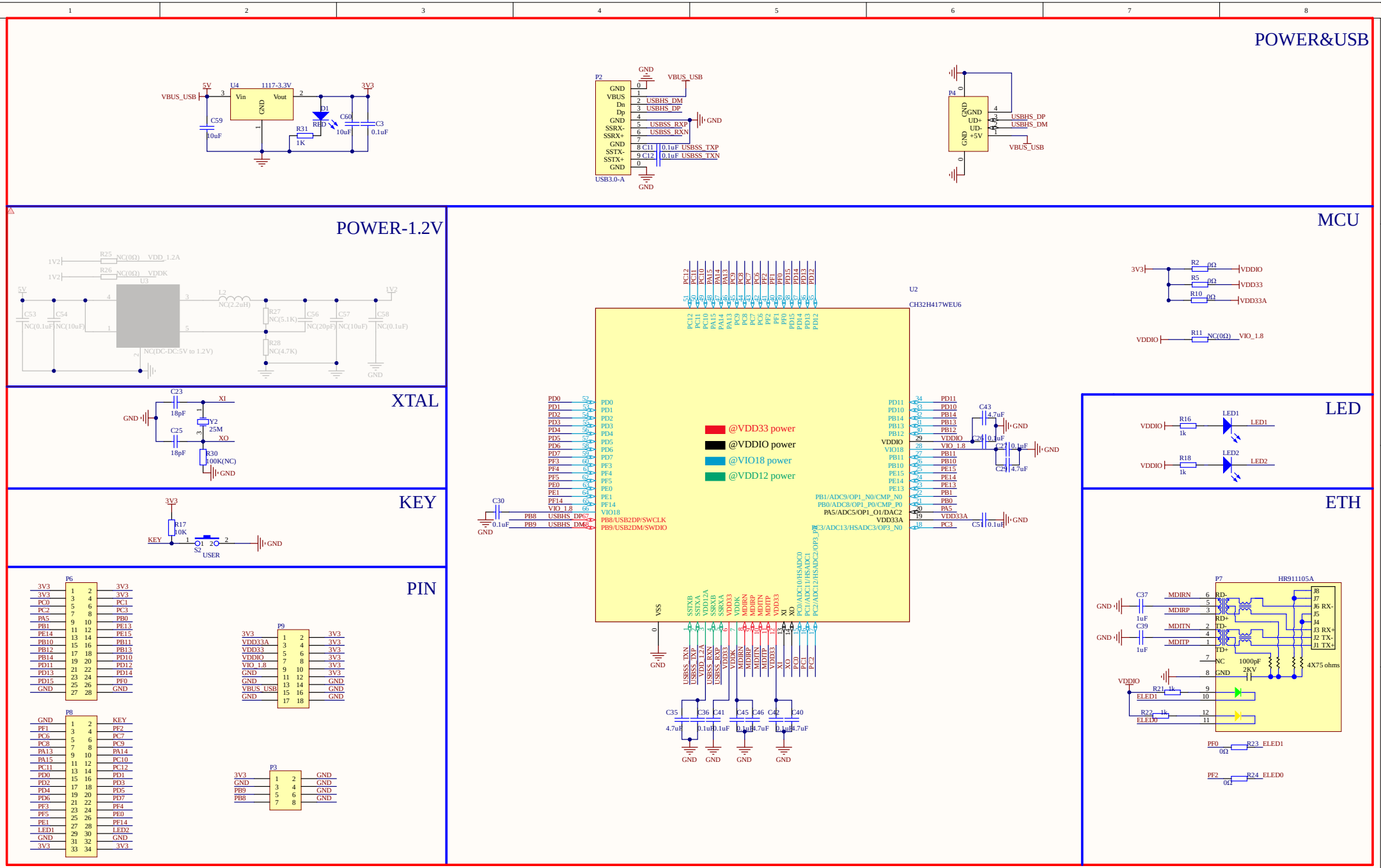
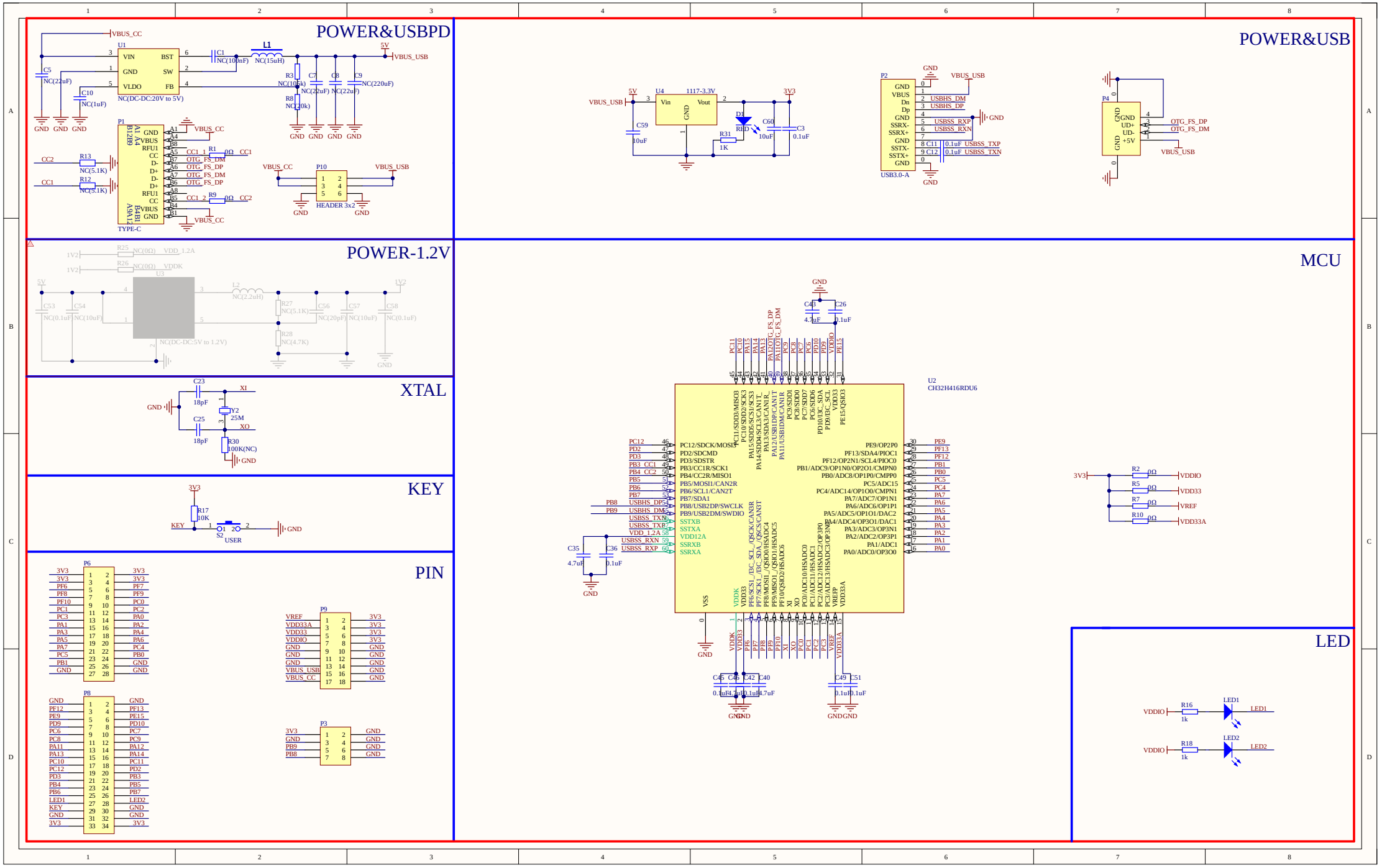

POWER&USBPD
U1
VIN
GND
VLDO
BST
SW
FB
NC(DC-DC:20V to 5V)
L1
NC(15uH)
VBUS_CC
VBUS_USB
5V
P1
TYPE-C
CC1
CC2
R13
NC(5.1K)
R12
NC(5.1K)
R1
0Ω
R9
0Ω
P10
HEADER 3x2
POWER-1.2V
U3
NC(DC-DC:5V to 1.2V)
L2
NC(2.2uH)
XTAL
Y1
32.768k
Y2
25M
R14
NC(0Ω)
R15
NC(0Ω)
R30
100K(NC)
KEY
S2
USER
RST
S1
RESET
PIN
P6
P5
P9
P8
P3
POWER&USB
U4
1117-3.3V
Vin
Vout
GND
D1
RED
P2
USB3.0-A
USBHS_DM
USBHS_DP
USBSS_RXP
USBSS_RXN
USBSS_TXP
USBSS_TXN
P4
OTG_FS_DP
OTG_FS_DM
MCU
U2
CH32H417QEU6
@VDD33 power
@VDDIO power
@VIO18 power
@VDD12 power
@VDD33&VBAT power
LED
LED1
LED2
R16
1k
R18
1k
ETH
P7
HR911105A
4X75 ohms
1000pF
2KV
R21
1k
R22
1k
R23_ELED1
R24_ELED0

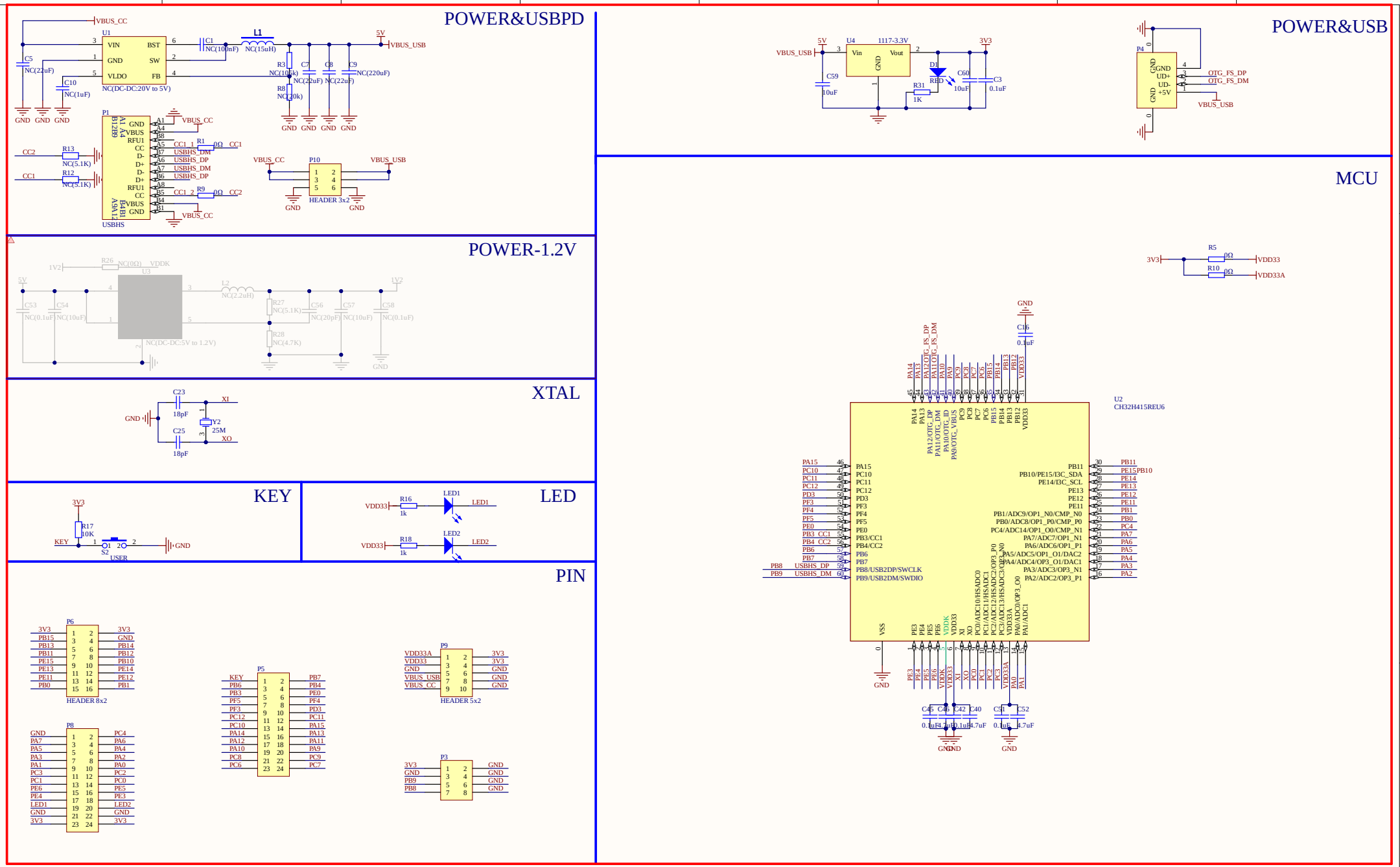

POWER&USBPD
U1
VIN
GND
VLDO
BST
SW
FB
NC(DC-DC:20V to 5V)
L1
NC(15uH)
C1
NC(100nF)
C5
NC(22uF)
C10
NC(1uF)
R3
NC(10k)
R8
NC(20k)
C7
NC(22uF)
C8
NC(22uF)
C9
NC(220uF)
5V
VBUS_USB
VBUS_CC
P1
TYPE-C
R13
NC(5.1K)
R12
NC(5.1K)
R1
0Ω
R9
0Ω
CC1_1
CC1_2
OTG_FS_DM
OTG_FS_DP
P10
HEADER 3x2
POWER&USB
U4
1117-3.3V
Vin
Vout
C59
10uF
D1
RED
R31
1K
C60
10uF
C3
0.1uF
3V3
P2
USB3.0-A
USBHS_DM
USBHS_DP
USBSS_RXP
USBSS_RXN
C11
0.1uF
USBSS_TXP
C12
0.1uF
USBSS_TXN
P4
OTG_FS_DP
OTG_FS_DM
POWER-1.2V
U3
NC(DC-DC:5V to 1.2V)
R25
NC(0Ω)
VDD_1.2A
R26
NC(0Ω)
VDDK
L2
NC(2.2uH)
R27
NC(5.1K)
R28
NC(4.7K)
C53
NC(0.1uF)
C54
NC(10uF)
C56
NC(20pF)
C57
NC(10uF)
C58
NC(0.1uF)
1V2
MCU
U2
CH32H417MEU6
R2
0Ω
R5
0Ω
R10
0Ω
VDDIO
VDD33
VDD33A
R11
NC(0Ω)
VIO_1.8
C17
0.1uF
C19
0.1uF
C21
0.1uF
C30
0.1uF
C35
4.7uF
C36
0.1uF
C50
4.7uF
C26
0.1uF
C27
0.1uF
C29
4.7uF
C44
0.1uF
C45
0.1uF
C46
4.7uF
C47
0.1uF
C43
4.7uF
C49
0.1uF
C51
0.1uF
LED
R16
1k
LED1
R18
1k
LED2
KEY
S2
USER
R17
10K
RST
S1
RESET
C31
0.1uF
XTAL
C23
18pF
C25
18pF
Y2
25M
R30
100k(NC)
XI
XO
PIN
P6
P5
P3
P8
P9
P11
ETH
P7
HR911105A
C37
1uF
C39
1uF
MDIRN
MDIRP
MDITN
MDITP
1000pF
2KV
4X75 ohms
R21
1k
R22
1k
ELED1
ELED0
R23_ELED1
R24_ELED0
PF0
PF2
0Ω


USBSS
P1
USB3.0-A
GND
VBUS
Dn
Dp
SSRX-
SSRX+
SSTX-
SSTX+
USB_UD_n
USB_UD_p
USB_SSRX_p
USB_SSRX_n
USB_SSTX_p
USB_SSTX_n
F1
2A
C4
47uF
+5V
R5
1K
LED1
RED
C5
C6
0.1uF
POWER
U2
VIN
SW
RUN
FB/VOUT
8503
L1
2.2uH
R11
10K
C7
20pF
C8
10uF
C9
0.1uF
R12
2.2K
3V3
C10
C11
U3
L2
R13
10K
C16
C17
C18
R14
1V2
C21
C22
NC
SPI FLASH
U4
CS
D0
WP
GND
VCC
HOLD
CLK
DI
SPI FLASH
SPI_CS
SPI_MISO
SPI_SCK
SPI_MOSI
C29
DEBUG
P2
TX6
SWCLK
SWDIO
HEADER 6
LED2
BLUE
R9
470Ω
LED3
GREEN
R10
P3
HEADER 30x2
MCU
U1
CH32H417MEU6
R1
R2
R3
R6
R7
0R
VDDK
VDD12A
VDD33
VDD33A
VDDIO
C1
0.1uF
C2
C3
C13
C20
C19
4.7uF
C23
C24
20pF
Y1
25M
R15
NC
S1
NRST
C30
C32
C33
C34
C25
C26
C35
C27
C12
C14
C15
C28
@VDD33 power
@VDDIO power
@VIO18 power
@VDD12 power
PB3/CC1
PB4/CC2
PB8/USB2DP/SWCLK
PB9/USB2DM/SWDIO
PB1/ADC9/CMP_N0/OP1_N0
PB0/ADC8/OP1_P0/CMP_P0
PA5/ADC5/OP1_O1/DAC2
VSS


USBSS
P1
USB3.0-A
GND
VBUS
Dn
Dp
SSRX-
SSRX+
SSTX-
SSTX+
USB_UD_n
USB_UD_p
USB_SSRX_p
USB_SSRX_n
USB_SSTX_p
USB_SSTX_n
F1
2A
C4
47uF
+5V
R5
1K
LED1
RED
C5
C6
0.1uF
POWER
U2
U3
VIN
SW
RUN
FB/VOUT
8503
L1
2.2uH
L2
2.2uH
3V3
1V2
R11
10K
R12
2.2K
R13
10K
R14
10K
C7
20pF
C8
10uF
C9
0.1uF
C10
C11
C21
C22
C16
C17
C18
SPI FLASH
U4
CS
D0
WP
GND
VCC
HOLD
CLK
DI
SPI_CS
SPI_MISO
SPI_SCK
SPI_MOSI
C29
NC
DEBUG
P2
TX6
SWCLK
SWDIO
HEADER 6
LED2
BLUE
R9
470Ω
LED3
GREEN
R10
P3
HEADER 30x2
MCU
U1
CH32H417MEU6
@VDD33 power
@VDDIO power
@VIO18 power
@VDD12 power
R1
R2
R3
R6
R7
0R
VDDK
VDD12A
VDD33
VDD33A
VDDIO
C1
0.1uF
C2
C3
C12
C14
4.7uF
C15
C28
C13
C20
C19
C23
C24
20pF
Y1
25M
R15
NC
S1
NRST
C30
C32
C33
C34
C25
C26
C35
C27
VSS


POWER&USB
U4
1117-3.3V
Vin
Vout
GND
5V
3V3
VBUS_USB
C59
10uF
D1
RED
C60
10uF
C3
0.1uF
R31
1K
P2
USB3.0-A
USBHS_DM
USBHS_DP
USBSS_RXP
USBSS_RXN
0.1uF USBSS_TXP
0.1uF USBSS_TXN
P4
USBHS_DP
USBHS_DM
VBUS_USB
POWER-1.2V
U3
VDD_1.2A
VDDK
L2
NC(2.2uH)
NC(DC-DC:5V to 1.2V)
1V2
XTAL
C23
18pF
C25
18pF
Y2
25M
XI
XO
R30
100K(NC)
KEY
R17
10K
S2
USER
PIN
P6
P9
P8
P3
MCU
U2
CH32H417WEU6
@VDD33 power
@VDDIO power
@VIO18 power
@VDD12 power
PB8/USB2DP/SWCLK
PB9/USB2DM/SWDIO
PA5/ADC5/OP1_O1/DAC2
PB1/ADC9/OP1_N0/CMP_N0
PB0/ADC8/OP1_P0/CMP_P0
PC3/ADC13/HSADC3/OP3_N0
C30
0.1uF
C43
4.7uF
C26
0.1uF
C27
0.1uF
C29
4.7uF
C51
0.1uF
R2
0Ω
R5
0Ω
R10
0Ω
R11
NC(0Ω)
VIO_1.8
LED
R16
1k
R18
1k
LED1
LED2
ETH
P7
HR911105A
C37
1uF
C39
1uF
MDIRN
MDIRP
MDITN
MDITP
1000pF
2KV
4X75 ohms
R21
1k
R22
1k
ELED1
ELED0
R23_ELED1
R24_ELED0
0Ω


POWER&USBPD
U1
VIN
GND
VLDO
BST
SW
FB
NC(DC-DC:20V to 5V)
L1
NC(15uH)
C1
NC(100nF)
C5
NC(22uF)
C10
NC(1uF)
R3
NC(100k)
R8
NC(10k)
C7
NC(22uF)
C8
NC(22uF)
C9
NC(220uF)
5V
VBUS_USB
VBUS_CC
P1
TYPE-C
CC2
CC1
R13
NC(5.1K)
R12
NC(5.1K)
R1
0Ω
R9
0Ω
P10
HEADER 3x2
POWER&USB
U4
1117-3.3V
Vin
Vout
3V3
C59
10uF
D1
RED
R31
1K
C60
10uF
C3
0.1uF
P2
USB3.0-A
USBHS_DM
USBHS_DP
USBSS_RXP
USBSS_RXN
USBSS_TXP
USBSS_TXN
0.1uF
P4
OTG_FS_DP
OTG_FS_DM
POWER-1.2V
U3
NC(DC-DC:5V to 1.2V)
L2
NC(2.2uH)
R25
NC(0Ω)
VDD_1.2A
R26
NC(0Ω)
VDDK
C53
NC(0.1uF)
C54
NC(10uF)
R27
NC(5.1K)
C56
NC(20pF)
C57
NC(10uF)
C58
NC(0.1uF)
R28
NC(4.7K)
1V2
XTAL
C23
18pF
C25
18pF
X1
XO
Y2
25M
R30
100K(NC)
KEY
R17
10K
S2
USER
MCU
U2
CH32H416RDU6
C43
4.7uF
C26
0.1uF
C35
4.7uF
C36
0.1uF
C45
0.1uF
C46
4.7uF
C42
0.1uF
C40
4.7uF
C49
0.1uF
C51
0.1uF
R2
0Ω
R5
0Ω
R7
0Ω
R10
0Ω
VDDIO
VDD33
VREF
VDD33A
PIN
P6
P8
P9
P3
LED
LED1
LED2
R16
1k
R18
1k


POWER&USBPD
U1
VIN
GND
VLDO
BST
SW
FB
NC(DC-DC:20V to 5V)
C1
NC(100nF)
L1
NC(15uH)
5V
VBUS_USB
VBUS_CC
C5
NC(22uF)
C10
NC(1uF)
R3
NC(100k)
R8
NC(10k)
C7
NC(22uF)
C8
NC(22uF)
C9
NC(220uF)
GND
P1
USBHS
CC2
R13
NC(5.1K)
CC1
R12
NC(5.1K)
R1
0Ω
R9
0Ω
USBHS_DM
USBHS_DP
P10
HEADER 3x2
POWER&USB
U4
1117-3.3V
Vin
Vout
3V3
C59
10uF
D1
RED
R31
1K
C60
10uF
C3
0.1uF
P4
OTG_FS_DP
OTG_FS_DM
POWER-1.2V
U3
1V2
R26
NC(0Ω)
VDDK
5V
C53
NC(0.1uF)
C54
NC(10uF)
NC(DC-DC:5V to 1.2V)
L2
NC(2.2uH)
R27
NC(5.1K)
R28
NC(4.7K)
C56
NC(20pF)
C57
NC(10uF)
C58
NC(0.1uF)
MCU
R5
0Ω
R10
0Ω
VDD33
VDD33A
C16
0.1uF
U2
CH32H415REU6
XTAL
C23
18pF
C25
18pF
Y2
25M
XI
XO
KEY
R17
10K
S2
USER
LED
R16
1k
LED1
R18
1k
LED2
PIN
P6
HEADER 8x2
P5
P9
HEADER 5x2
P8
P3